

Contaminants Control Procedure

Process 1	Identification of key control products	Personnel resource	No designation
		Material resource	No designation

INPUT

The suppliers shall designate the products as subject to intensive control of contaminants if the products apply to the following items.

- The products whose cleanliness ranks and contaminant mass criteria are specified on the drawing in accordance with HES (Honda Engineering Standards): see Table 1 and 2
- Products subject to “Contaminants Control Products (automatic transmission, etc.)”: see Table 9, pp4
- Products subject to “Contaminants Control Products (other than automatic transmission, etc.)”: see Table 10, pp5

In addition, the suppliers shall identify the cleanliness rank and the management importance rank if the product is subject to intensive control of contaminants.

Table 1 Definition of cleanness rank (automatic transmission, etc.)

Cleanness Rank	Description
A	Control circuit parts whose failure, such as valve lock failure by contact with contaminants, causes to lose function and quality of drive train.
B	Parts having contact with hydraulic fluid excluding control circuit parts whose failure, such as valve lock by contact with contaminants, causes to lose function and quality of drive train.
C	Parts not included in rank A nor B which can contaminate parts in rank A or B indirectly by contact with contaminants.

Table 2 Definition of management importance rank (other than automatic transmission, etc.)

Management importance rank	Description
A	Parts which may seriously injure basic functions (run, turn, stop, etc.) of products by contaminants.
B	Parts which may affect basic functions (run, turn, stop, etc.) of products by contaminants.

Process 2	Designation of section and responsible person for contaminant control	Personnel resource	No designation
		Material resource	No designation

INPUT

The suppliers shall appoint a section and person responsible for control items, control points, etc., shown in Table 3 in order to maintain the products subject to contaminants control in a manner satisfactory to contaminant requirements until delivered to Honda.

Table 3 Control items and control points of products subject to contaminants control

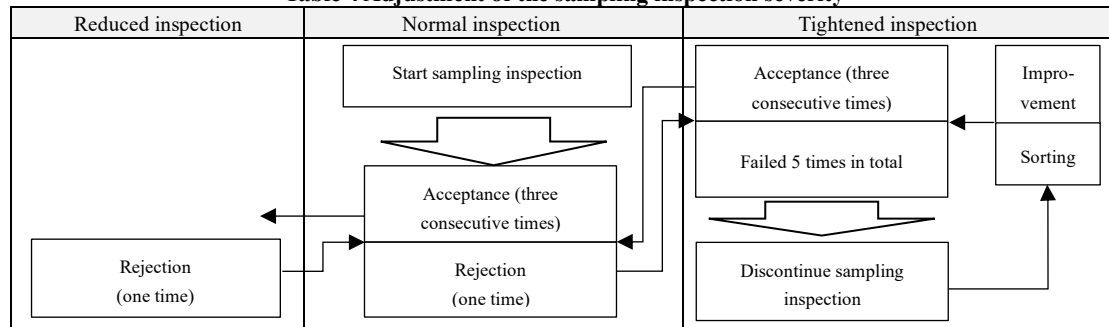
No.	Control Item	Control Pint									
1	Design of parts	1. Placing of cleanness ranking and contaminant mass criteria in accordance with HES A3054 “DESIGNATION FOR CLEANLINESS OF AUTOMATIC TRANSMISSION”.									
2	Verification of cleanliness ranks and management importance ranks	1. Verification <table border="1" style="width: 70%; border-collapse: collapse;"> <tr> <td style="width: 30%;"></td> <td style="width: 20%; text-align: center;">Automatic transmission, etc.</td> <td style="width: 50%; text-align: center;">Other than automatic transmission, etc.</td> </tr> <tr> <td style="text-align: center;">Cleanness rank</td> <td> • Drawing • Table 9 Contaminants Control Products (automatic transmission, etc.) </td> <td style="text-align: center;">—</td> </tr> <tr> <td style="text-align: center;">Management priority rank</td> <td style="text-align: center;">—</td> <td> • Table 10 Contaminants Control Products (other than automatic transmission, etc.) </td> </tr> </table>		Automatic transmission, etc.	Other than automatic transmission, etc.	Cleanness rank	• Drawing • Table 9 Contaminants Control Products (automatic transmission, etc.)	—	Management priority rank	—	• Table 10 Contaminants Control Products (other than automatic transmission, etc.)
	Automatic transmission, etc.	Other than automatic transmission, etc.									
Cleanness rank	• Drawing • Table 9 Contaminants Control Products (automatic transmission, etc.)	—									
Management priority rank	—	• Table 10 Contaminants Control Products (other than automatic transmission, etc.)									
3	Setting of contaminants control plans	1. Define items subject to contaminants control items during processing and transportation. 2. Arrange preventives against contaminants mix-up. 3. Reflect contaminants control items in the standard documents.									
4	Setting up of Delivery Packaging	1. Define packaging specifications and delivery packaging with preventive measures for contaminants mix-ups.									
5	Processing of parts	1. Cleaning control of processing facilities, jig & fixture, and work supplies. 2. Control of working conditions in the process of cleaning parts. 3. Cleanliness control of circulating oil from QA device to parts.									
6	Inspection of contaminants	1. Setting up of inspection method and feedback of inspection results. 2. Agreement on contaminants control with supplier. 3. Instruction for improvement of suppliers' contaminants control activities.									
7	Release/delivery	1. Implementation of preventive measures for keeping contaminants from entering release/delivery parts. 2. Agreement on contaminants control with outside contractors for transportation.									
8	Verification of delivery package	1. Feedback of delivery package verification.									
9	Storage of products, components	1. Ensure prevention of contaminants in the storage products, components.									

Process 3	Inspection method for contaminants	Personnel resource	Qualified inspector
		Material resource	Measurement equipment used for contaminant inspection (including microscope, magnifier, filter paper, etc.)

INPUT

The suppliers shall conduct a sampling-based contaminants inspection of the products subject to intensive control, taking into consideration the following items.

- a) Adjustment of the sampling inspection severity (start from normal inspection: see Table 4)

Table 4 Adjustment of the sampling inspection severity

- b) The inspection frequency for each inspection severity category (see Table 5)

Table 5 Inspection frequency

Severity of inspection	Reduced inspection	Normal inspection	Tightened inspection
Frequency of inspection	Once a month	Once every two weeks	Once a week

- c) Number of samples according to products subject to contaminants control (see Table 6)

Automatic transmission, etc.: Number of samples in accordance with the product mass category
(if other than a whole number, shall be rounded)

Other than automatic transmission, etc.: Number of samples in accordance with each inspection category of severity

Table 6 Number of inspection samples

Automatic transmission, etc.	Classification by product mass*	Less than 10g	11 ~ 100g	101 ~ 500g	501 ~ 1000g	More than 1001g
	Sample size	10	5	5	3	3
Other than automatic transmission, etc.	Severity of inspection	Reduced inspection		Normal inspection		Tightened inspection
	Sample size	2		3		3

* Product mass: see the values indicated on drawings

- d) The criteria for inspection of contaminants according to products subject to contaminants control (see Table 7)
Automatic transmission, etc.: Product mass criteria shown in **Table 7**, and the size of contaminants shall be 0.4mm MAX. (as a target value)

However, the suppliers who deliver products to Dongfeng Honda Automobile Co., Ltd (WDHAC) shall set the target size of contaminants at 0.1 mm or less.

(See SQM WDHAC E 部品附则 第 3 章 过程设计与开发 3.7 工序质量控制表 (II)/控制计划)

Other than automatic transmission, etc.: Size and quantity of contaminants shown in **Table 8**

**Table 7 Judgment criteria of contaminants inspection for automatic transmission, etc.
(allowable maximum mass of contaminants per product)**

Product mass	10g or less	11 ~ 100g	101 ~ 500g	501 ~ 1000g	1001g and more
Cleanliness rank					
A	0.5mg or less	0.7mg or less	1.0mg or less	1.1mg or less	1.3mg or less
B	0.7mg or less	1.0mg or less	1.4mg or less	1.6mg or less	1.8mg or less
C	1.4mg or less	2.0mg or less	2.8mg or less	3.2mg or less	3.6mg or less

Table 8 Judgment criteria of contaminants inspection for other than automatic transmission, etc.

Management priority rank	Criteria
A	No contaminants of about 0.2 mm or larger in size shall be found. <u>The suppliers who deliver products to Dongfeng Honda Automobile Co., Ltd (WDHAC) shall ensure that the criteria of contaminants inspection is that there are no contaminants of 0.2 mm or larger in size.</u> <u>(See SQM WDHAC E 部品附则 第3章 过程设计与开发 3.7 工序质量控制表 (II)/控制计划)</u>
B	Quantity of contaminants of about 0.2 mm shall be six or less, and also no contaminants of 0.2mm or larger in size shall be found.

e) The criteria for inspection of contaminants according to products subject to contaminants control including the following items

Automatic transmission, etc.	1. Collection of cleaning agent used for the inspection sample in accordance with part mass category 2. Filtration of the cleaning agent using filter paper with a retained particle size equivalent to 1 μm 3. Measurement of the total mass of collected contaminants and calculation of the mass of contaminants per test sample 4. Measurement of the size of the collected contaminants using magnifying lenses, microscopes, etc. 5. Only if the size and quantity of contaminants of products satisfy the judgment criteria, the products shall be deemed acceptable
Other than automatic transmission, etc.	1. Collection of cleaning agent used for the inspection sample in each inspection sample unit 2. Filtration of the cleaning agent using filter paper 3. Measurement of the size of the collected contaminants using magnifying lenses, microscopes, etc. 4. Only if the size and quantity of contaminants of products satisfy the judgment criteria, the products shall be deemed acceptable

Process 4	Control of inspection records, etc.	Personnel resource	No designation
		Material resource	No designation

INPUT

Collected contaminants and inspection records of the collected contaminants shall be controlled in a manner that allows the supplier to present the contaminants or records to Honda whenever so requested by Honda.

Table 9 Contaminants Control Products (automatic transmission, etc.)

Category of Mechanism	Cleanliness rank							
	A			B			C	
AT	TC case COMP Transmission case R side cover COMP Clutch ASSY Feed pipe COMP Main shaft COMP Counter shaft COMP Secondary shaft Shift rail Oil pump gear ATF strainer TRQ/CONV ASSY Main valve body ASSY Separate plate Shift valve cover	Regulator valve body ASSY Governor ASSY Servo body ASSY Linear solenoid ASSY Accum. piston L/C body ASSY Secondary body ASSY Secondary filter Second accum. body ASSY MOD body ASSY Shift solenoid ASSY L/C solenoid ASSY Oil pressure sensor Stator shaft COMP	Baffle plate COMP Breather pipe Breather cap Transfer case Transfer side cover T case packing R side cover packing Oil guide cover Oil guide plate LOW gear Counter LOW gear One-way clutch ASSY Parking gear M2gear C2gear	C3 gear M4 gear C4gear Counter reverse gear Reverse idle shaft holder COMP Reverse idle gear Reverse gear selector Selector hub Speed meter gear ASSY Speed meter gear holder Gear holder seal Speed meter gear COMP Companion flange COMP Reverse shift fork Servo detent base	Shift rail screw set Servo detent ASSY Control shaft COMP Manual valve pin Manual valve roller Change shaft COMP Servo detent sleeve COMP P/BRK P/BRK spring P/BRK stopper Parking pole spring P/BRK pole Parking shaft Cooler hose Transmission magnet COMP	Suction pipe Oil cooler (including trailer traction kit) ATF gauge COMP Oil level pipe COMP ATF filter bolt Oil hose pipe COMP Joint bolt COMP Joint bolt Bypass body Bypass body cover Throttle valve shaft COMP Throttle cam stopper Throttle cable stay Accum. cover Accum. spring	Pick up ASSY Drive transfer gear Differential ASSY Hypoid pinion Hypoid ring gear Final driven gear Throttle adjust bolt Extension shaft	Transmission hanger Torque converter cover Change cover Transmission under guard Speed meter gear holder plate Throttle cable COMP Control lever Control wire holder Throttle control lever Position sensor ASSY Meter cable clip Oil pipe clip COMP
CVT	Fly wheel case Transmission case COMP R side cover ASSY Forward clutch ASSY RVS brake piston Start clutch ASSY Feed pipe Oil path pipe COMP	Manual valve body pipe COMP Input shaft COMP Oil pump ASSY ATF Filter COMP Valve body ASSY Manual valve body ASSY Manual separate plate Linear solenoid ASSY	Oil pan Differential cover Intermediate plate ASSY Pulley cover plate COMP A FW case packing I/M plate packing R over packing Oil pan packing	RVS return spring COMP RVS brake plate RVS brake disc RVS brake disc spring RVS brake end plate Oil path pipe holder COMP LUB pipe Secondary gear shaft	Secondary drive gear ASSY Planetary gear COMP Sun gear Ring gear COMP Drive sprocket hub Control lever Control shaft COMP P/BRK rod holder	Parking brake pole Parking pole SPG Detent spring ASSY O/P drive sprocket O/P drive chain Pitot pipe Pitot LUB pipe Pitot flange	Pitot pipe stay ATF gauge COMP Oil hose pipe COMP B Pick up ASSY Final driven gear Oil pipe clip COMP Cotter retainer Circlip retainer	Transmission hanger Control wire stay
4WD	Pump body ASSY Clutch piston	ATF strainer D ring	Rear differential magnet COMP Companion flange Differential ASSY Differential carrier SET	TCD case Hypoid gear Pinion spacer Piston spring	Oil pump pin Back-up ring Pinion gear SET Differential case	Pinion shaft Transfer shaft Driven transfer gear Transfer drive bevel gear	Transfer driven bevel gear Differential clutch ASSY Oil pump drive shaft Transfer case Transfer side cover	Breather tube joint Breather tube clamp Breather tube ASSY Breather tube
MCU	R/L clutch case COMP L side cover ASSY Solenoid valve body Solenoid separate plate Linear solenoid ASSY Shift solenoid ASSY Oil pressure sensor Oil TEMP sensor	Regulator valve Oil pump driven gear Oil pump drive gear Clutch piston ATF strainer COMP Regulator valve spring Regulator valve cap	R side cover R side cover packing Breather cover packing Extension tube COMP Extension shaft	Center shaft R/L clutch case packing Solenoid body packing Breather cover COMP Valve cap plate	Planetary ASSY L central gear R central joint COMP R central gear C central gear	Clutch disc Clutch plate Clutch return spring Spring guide plate Clutch end plate	Special bolt Oil pump gear cover Oil hose pipe COMP	Under cover MCU sub-harness Harness clamp stay Sub-harness stay Breather tube ASSY
Common parts	Seal ring		Cooler hose Drive shaft Half shaft Flange bolt O ring Cotter	Thrust washer Distance collar Snap ring Ball bearing Ball radial bearing Taper bearing	Needle bearing Thrust needle bearing Oil seal Collar Roller circlip	Washer Shim Snap ring Dowel pin Hex nut Flange nut	Spring washer Drain plug bolt Plug washer Differential pinion thrust washer Spring pin Radiator (AT oil cooler)	

Table 10 Contaminants Control Products (other than automatic transmission, etc.)

Management importance rank	A	B
Part group	◆ Brake system • Brake master cylinder • ABS solenoid valve • ABS ASSY • Brake hose (pipe) • Wheel cylinder • Caliper ASSY • Master power ASSY • Proportioning valve ASSY ◆ Steering system • Steering gear BOX • P/S pump ASSY • Speed sensor • Oil tank • Feed hose • Pipe, tube • Oil cooler ◆ Fuel system • Carburetor ASSY • Fuel feed pipe COMP • Fuel strainer COMP • Circuit component filtered through fuel filter	◆ Driving channel system • Water pump COMP • Radiator ASSY • Water hose • Reserve tank • Cooling water circuit component ◆ Lubrication system • Oil pump ASSY • Oil pressure switch • Manual transmission ASSY • Crank shaft COMP • Cylinder head • Cylinder block • Component with ENG ◆ Function system • Front fork • Front damper • Rear damper • Distributor • Completed system function part with rotating/sliding function ◆ EM control system • Control BOX ASSY • Rubber hose • EM control circuit component ◆ Fuel system • Fuel tank • Fuel pump • Circuit component not filtered through fuel filter